

The cross-entropy loss function is shown below,

$$L = -[y \cdot \log(a) + (1-y) \log(1-a)]$$

here, also the chain rule applicable and the only change is for dL/da . the other

term $\frac{da}{dz} = a(1-a)$ ~~is the same as the~~

previous sigmoid function ($\sigma(z) = \frac{1}{1+e^{-z}}$).

Now apply $\frac{d}{dn}(a+b) = \frac{d}{dn}(a) + \frac{d}{dn}(b)$.

$$\frac{d}{da}(-y \log(a)) = -y \cdot \left(\frac{1}{a}\right) = \frac{-y}{a}$$

$$\frac{d}{da}((1-y) \log(1-a)) = \frac{-(1-y)}{1-a} \cdot (-1) = \frac{(1-y)}{(1-a)}$$

$$\therefore \frac{dL}{da} = -y/a + (1-y)/(1-a)$$

$$\text{finally } \frac{dL}{da} = (a-y)/(a(1-a))$$

$$\text{Now, } \sigma = \frac{dL}{dz} = \frac{dL}{da} \cdot \frac{da}{dz} = \left[\frac{(a-y)}{a(1-a)} \right] \cdot (a(1-a)) = a-y$$